

UPDATED DATA DICTIONARY

1. Introduction

This updated data dictionary describes the variables included in the final machine-learning ready loan prediction dataset produced at the end of week 3. The dataset was refined using the findings from advanced exploratory data analysis, statistical analysis, feature engineering and feature evaluation.

The final dataset contains:

- 614 observations,
- 23 variables,
- 22 predictor variables,
- 1 target variable,
- 0 missing values.

Categorical variables were encoded into numerical binary variables, while numerical predictors were scaled where applicable.

2. Final dataset variables

Variable	Type	Processing	Description
ApplicantIncome	Numerical	Standardized	Income of the primary loan applicant.
CoapplicantIncome	Numerical	Standardized	Income contributed by the coapplicant.
LoanAmount	Numerical	Standardized	Amount of the loan requested by the applicant.
Loan_Amount_Term	Numerical	Standardized	Term of the requested loan.
Credit_History	Numerical/Binary	Retained	Indicates the applicant's credit-history status.
TotalIncome	Numerical	Standardized	Combined income of the applicant and coapplicant.

DebtToIncomeRatio	Numerical	Standardized	Engineered measures represent the relationship between loan amount and income.
Gender_Male	Binary	One-hot encoded	Indicates whether the applicant belongs to the Male category.
Married_Yes	Binary	One-hot encoded	Indicates whether the applicant is married.
Dependents_1	Binary	One-hot encoded	Indicates one dependent.
Dependents_2	Binary	One-hot encoded	Indicates two dependents.
Dependents_3+	Binary	One-hot encoded	Indicates three or more dependents.
Education_Not Graduate	Binary	One-hot encoded	Indicates that the applicant is not a graduate.
Self_Employed_Yes	Binary	One-hot encoded	Indicates that the applicant is self-employed.
Property_Area_Semiurban	Binary	One-hot encoded	Indicates a semiurban property area.
Property_Area_Urban	Binary	One-hot encoded	Indicates an urban property area.
IncomeBand_Lower-Middle	Binary	One-hot encoded	Indicates the Lower-Middle income category.
IncomeBand_Upper-Middle	Binary	One-hot encoded	Indicates the Upper-Middle income category.
IncomeBand_High	Binary	One-hot encoded	Indicates the High income category.
FamilySizeGroup_Medium Family	Binary	One-hot encoded	Indicates the Medium Family category.

FamilySizeGroup_Small Family	Binary	One-hot encoded	Indicates the Small Family category.
FamilySizeGroup_Unknown	Binary	One-hot encoded	Represents observations where the family-size group was classified as Unknown.
Loan_Status	Binary Target	Encoded	Target variable indicating loan approval: 1 = Approved and 0 = Rejected.

3. Numerical variables

The final modelling dataset contains seven numerical predictor variables:

- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- TotalIncome
- DebtToIncomeRatio

The continuous numerical predictors were scaled during the final modelling preparation process. Scaling ensures that numerical variables with different ranges are placed on a comparable scale, which is particularly useful for machine-learning algorithms that are sensitive to feature magnitude.

4. Encoded categorical variables

Categorical variables were transformed into binary numerical variables through one-hot encoding.

The encoded variables include:

Gender

Gender_Male

Married Status

Married_Yes

Dependents

- Dependents_1
- Dependents_2
- Dependents_3+

Education

Education_Not Graduate

Self-Employment

Self_Employed_Yes

Property Area

- Property_Area_Semiurban
- Property_Area_Urban

Income Band

- IncomeBand_Lower-Middle
- IncomeBand_Upper-Middle
- IncomeBand_High

Family Size Group

- FamilySizeGroup_Medium Family
- FamilySizeGroup_Small Family
- FamilySizeGroup_Unknown

One category from each original categorical variable serves as the reference category and is therefore not represented by a separate encoded column.

5. Target variable

The target variable is:

Loan_Status

It was encoded numerically as:

Value	Meaning
1	Loan Approved
0	Loan Rejected

The final target distribution is:

Loan status	count
Approved (1)	422
Rejected (0)	192

The target variable is separated from the predictor variables during machine-learning model development.

6. Excluded variables

Two variables that appeared during week 3 analysis were excluded from the final modelling dataset.

Loan_ID

Loan_ID was removed because it is an identification field and does not contain meaningful information about the applicant or loan application.

CreditRiskCategory

CreditRiskCategory was removed because it was derived from Credit_History. Although it demonstrated a strong statistical association with Loan_Status, retaining it alongside Credit_History would introduce redundant information. Therefore, the original Credit_History variable was retained and CreditRiskCategory was excluded.

7. Final dataset structure

The final modelling dataset has the following structure:

Rows: 614

Columns: 23

Predictors: 22

Target: 1

Missing values: 0

The final dataset therefore contains a complete numerical representation of the predictors and target required for machine-learning modelling.

8. Final data quality validation

The final validation confirmed:

Dataset shape: (614, 23)

Total missing values: 0

Loan_Status missing values: 0

The target variable was also validated:

Loan_Status

1 422

0 192

This confirms that the final modelling dataset contains no missing target values and that all observations have a valid loan approval outcome.

9. Final modelling dataset

The final processed dataset is saved as:

final_modelling_dataset.csv

This dataset represents the final output of the week 3 dataset-refinement process and will serve as the input dataset for predictive modelling during week 4.

10. Conclusion

The updated data dictionary documents the final variables, their data types, processing methods and meanings. The final dataset has been refined through advanced exploratory analysis, statistical analysis, feature engineering and feature evaluation. Irrelevant and redundant variables were removed, categorical variables were encoded, numerical predictors were scaled, and the target variable was encoded for classification modelling. The resulting 614 × 23 dataset contains 22 predictors and one target variable, with zero missing values, and is ready for the machine-learning stage of the project.